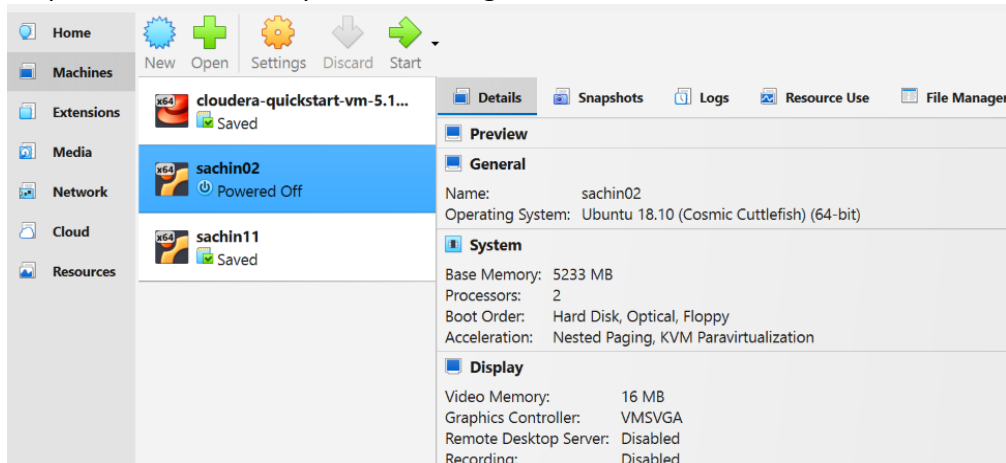


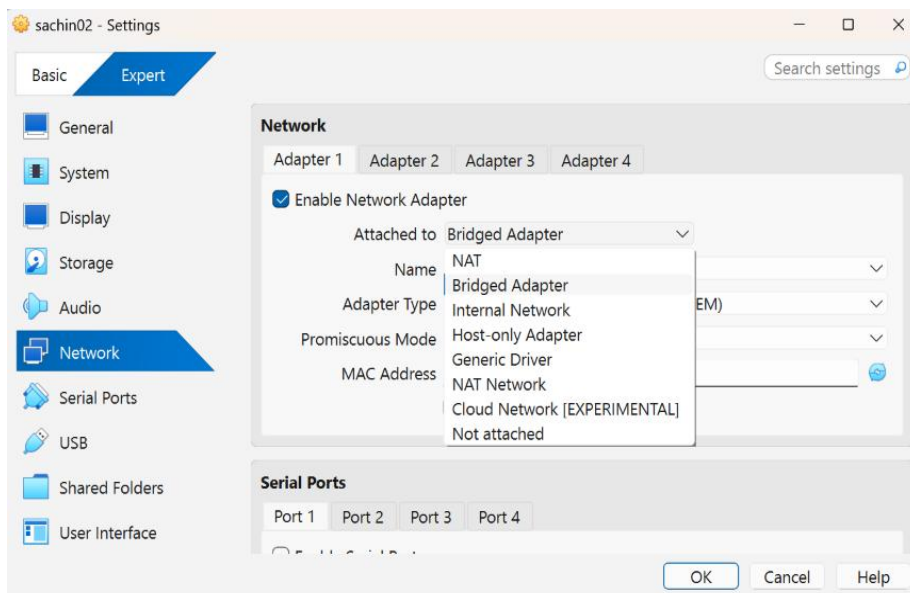
Practical – 7

AIM:- How to configure bridged network between virtual machines in virtual box.

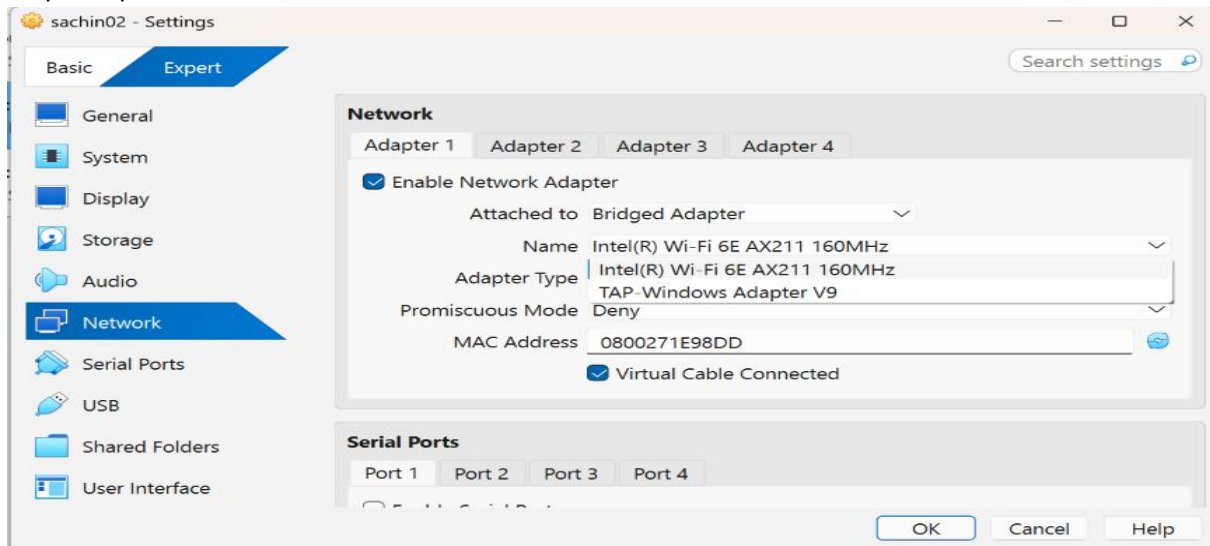
Step 1: In VirtualBox, open the settings for 'virtual machine'.



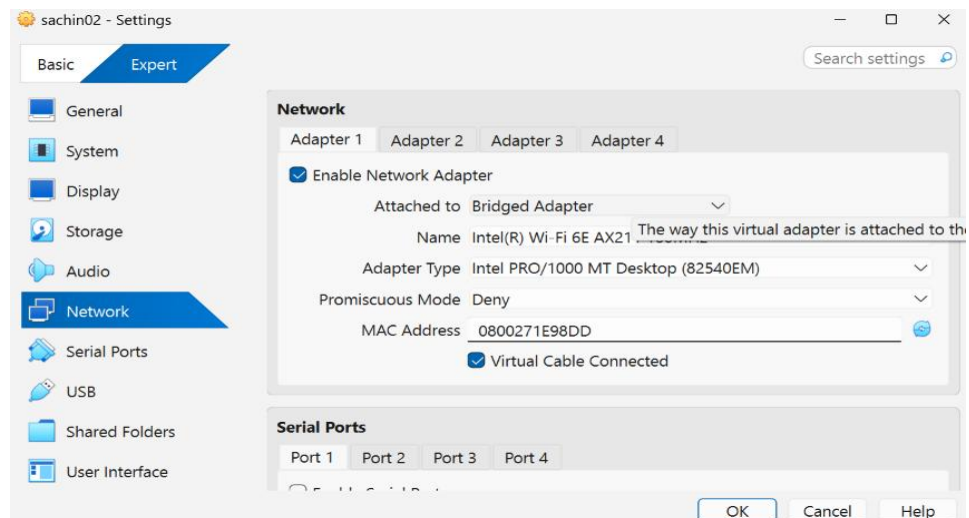
Step 2: Within the settings, navigate to the network configuration and modify the "Attached to" option from "Internal Network" to "Bridged Adapter. "



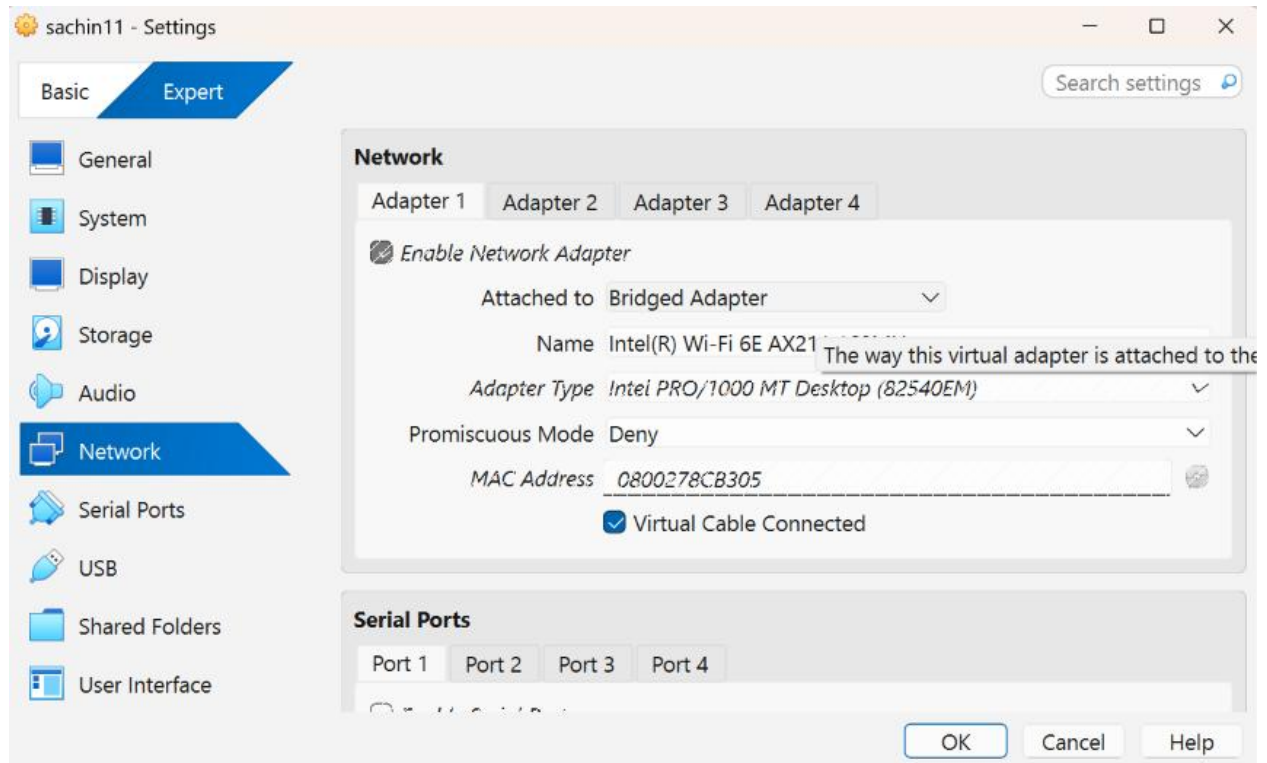
Step 3: Input Name for Wi-Fi or Ethernet Connection



4: Click OK after confirming all settings



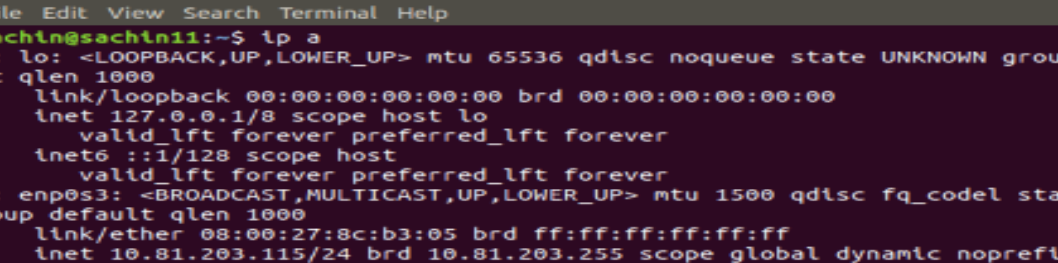
Step 5: Repeat the identical procedure for virtual machine 2.



Step 6: Begin both virtual machines. After they are running, open the terminal on each and run the ifconfig or ip a command.

The observed IP addresses are:● Machine 1 : 172.17.25.3Machine 2: 10.0.2.10

```
sachin@sachin02: ~  
sachin@sachin02:~$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defau  
lt qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host  
        valid_lft forever preferred_lft forever  
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g  
roup default qlen 1000  
    link/ether 08:00:27:1e:98:dd brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.10/24 brd 10.0.2.255 scope global dynamic noprefixroute  
enp0s3  
        valid_lft 3146sec preferred_lft 3146sec  
    inet6 fe80::63af:adfd:e27:46dd/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
sachin@sachin02:~$
```



```
Activities  Terminal  Tue 10:35  sachin@sachin11: ~
File Edit View Search Terminal Help
sachin@sachin11:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defau
lt qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
    link/ether 08:00:27:8c:b3:05 brd ff:ff:ff:ff:ff:ff
    inet 10.81.203.115/24 brd 10.81.203.255 scope global dynamic noprefixroute
    enp0s3
        valid_lft 3370sec preferred_lft 3370sec
    inet6 fe80::5b0e:ddf8:69cc:af45/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
sachin@sachin11:~$
```

Step 7: Verify connectivity between the two virtual machines using the ping command.

Execute the following command:

ping 172 . 17 . 25 . 35 (Replace with the actual IP address)

The screenshot shows a Linux terminal window with a dark background. The title bar at the top indicates the system is running on Tuesday, October 10, 2018, at 11:03 AM, and the user is 'sachin' on the host 'sachin11'. The terminal displays the output of the 'ifconfig' command for the 'enp0s3' interface, showing its IP address (10.81.203.194), netmask (255.255.255.0), and other details. Below this, the user runs a 'ping' command to 10.81.203.194, which shows 11 successful pings with varying times. The terminal session ends with the user pressing Ctrl+C, resulting in '[1]+ Stopped ping 10.81.203.194'.

```

sachin@sachin11: ~
File Edit View Search Terminal Help

inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
inet6 ::1/128 scope host
    valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:8c:b3:05 brd ff:ff:ff:ff:ff:ff
    inet 10.81.203.115/24 brd 10.81.203.255 scope global dynamic noprefixroute enp0s3
        valid_lft 3370sec preferred_lft 3370sec
    inet6 fe80::5b0e:ddf8:69cc:af45/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
sachin@sachin11:~$ ping 10.81.203.194
PING 10.81.203.194 (10.81.203.194) 56(84) bytes of data:
 64 bytes from 10.81.203.194: icmp_seq=1 ttl=64 time=2.71 ms
 64 bytes from 10.81.203.194: icmp_seq=2 ttl=64 time=3.22 ms
 64 bytes from 10.81.203.194: icmp_seq=3 ttl=64 time=3.88 ms
 64 bytes from 10.81.203.194: icmp_seq=4 ttl=64 time=2.44 ms
 64 bytes from 10.81.203.194: icmp_seq=5 ttl=64 time=3.42 ms
 64 bytes from 10.81.203.194: icmp_seq=6 ttl=64 time=1.72 ms
 64 bytes from 10.81.203.194: icmp_seq=7 ttl=64 time=1.81 ms
 64 bytes from 10.81.203.194: icmp_seq=8 ttl=64 time=2.41 ms
 64 bytes from 10.81.203.194: icmp_seq=9 ttl=64 time=2.59 ms
 64 bytes from 10.81.203.194: icmp_seq=10 ttl=64 time=3.97 ms
 64 bytes from 10.81.203.194: icmp_seq=11 ttl=64 time=2.61 ms
^C
[1]+  Stopped                  ping 10.81.203.194
sachin@sachin11:~$
  
```

Name:-Sachin kumar

Rollno:-11242593